

Econ5/Poli5D: Introduction to Social Data Analytics: Week 1 Part 1

Why take this class?

- ▶ There has been an explosion in data availability from
 - ▶ Governments
 - ▶ Businesses
 - ▶ Non-Profits
 - ▶ Citizens
- ▶ Better computing power and statistical techniques to analyze it
- ▶ Requires knowledge of social science, computer programming and statistics to be useful

Learning objectives and class goals

This course has three main goals

- ▶ Introduce you to interesting and important social science questions
- ▶ Show how data can be used to provide answers to these questions
- ▶ Give you tools to perform this analysis through Excel, Stata, and R

What we use to analyze data

- ▶ There are a number of software options when analyzing data
- ▶ We will focus on Excel (1 week), Stata (4 weeks) and R (5 weeks)
- ▶ Why these three?
 - ▶ Excel is very popular in many industries
 - ▶ Stata commonly used in social sciences and will likely be used in subsequent econometrics/statistics courses
 - ▶ R is popular in both academic and business

Example of a Normal Week

- ▶ 1. Introduce Empirical Application for the week (often highlighting work from someone at UCSD). Make sure to watch the interview videos on Canvas!
- ▶ 2. Learn various commands/functions in a given software
- ▶ 3. Use those newly learned techniques to analyze data from the empirical application of the week
- ▶ 4. Complete quizzes/labs to evaluate progress in understanding the content

Each week there is a “Plan of the Week” that discusses what assessments are due and when

Weekly Labs

- ▶ Labs involve working on a short assignment with your classmates, with the help of a TA
 - ▶ At the end of class, TA will go through the answers to the lab
- ▶ Each week there is a short quiz related to the lab that you need to fill out in order to receive credit for lab
- ▶ If you cannot attend a lab in a given week, you can still receive credit by finishing the lab on your own time and then filling out the quiz related to the lab assignment and turning in the complete lab

Grade Breakdown

- ▶ Lab (10 percent of grade)
- ▶ Quizzes (15 percent of grade)
- ▶ Midterm (25 percent) – In class on Wednesday May 3rd
- ▶ Final (50 percent) – On Wednesday June 14th (Location TBA)

Academic Integrity

Acceptable collaborations

- ▶ Can you remind me which function is appropriate here?
- ▶ Which lectures had the notes on regression?

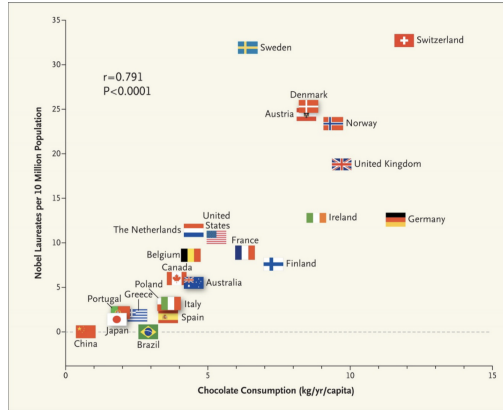
Unacceptable collaborations

- ▶ I can't figure out this one, can you send me your code

- ▶ Now that we've gone through the course structure, let's start the content so you can get a sense of how each week of the course will work
- ▶ This week we will be studying data from Brownback and Sadoff (2020)
- ▶ Brownback and Sadoff (2020) ran an experiment
- ▶ Before we get to the experiment, let's discuss why experiments are becoming popular in social science

Correlation vs. Causation

- ▶ Did you know that countries with higher consumption of chocolate per capita tend to win more Nobel prizes?



Correlation vs. Causation

- ▶ This is an example of a **correlation**
 - ▶ A **correlation** is any relationship between two variables, either by chance or causally
- ▶ Do we really think that chocolate consumption **causes** a country to win more Nobel prizes?

Correlation vs. Causation

- ▶ This is an example of a **correlation**
 - ▶ A **correlation** is any relationship between two variables, either by chance or causally
- ▶ Do we really think that chocolate consumption **causes** a country to win more Nobel prizes?
 - ▶ Probably not! Therefore, in this case, we have found a **spurious correlation**, or a correlation that arose due to chance

Spurious Correlations

- ▶ There are many spurious correlations out there
 - ▶ Go to <https://tylervigen.com/spurious-correlations> to see some of them!
- ▶ But social scientists and policymakers are often interested in inferring **causal relationships**
- ▶ If we need to decide if a given policy is worth it or not, we need to understand how it causally impacts individual's outcomes

Causality vs. Correlation

- ▶ Imagine a high school is trying to decide whether to invest in a new after-school program designed to help students with standardized testing prep

Causality vs. Correlation

- ▶ Imagine a high school is trying to decide whether to invest in a new after-school program designed to help students with standardized testing prep
- ▶ In the past, a subset of students had enrolled in a similar program offered outside of the school

Causality vs. Correlation

- ▶ Imagine a high school is trying to decide whether to invest in a new after-school program designed to help students with standardized testing prep
- ▶ In the past, a subset of students had enrolled in a similar program offered outside of the school
- ▶ In order to try and understand the benefits of the program, the school decides to use data on past students to see if those who attended SAT prep programs outperformed students that did not.

Causality vs. Correlation

- ▶ The school finds those who attended test prep programs score much higher on standardized tests than those who did not
- ▶ Does this mean the test prep program works and the school should invest in one?

Causality vs. Correlation

- ▶ The school finds those who attended test prep programs score much higher on standardized tests than those who did not
- ▶ Does this mean the test prep program works and the school should invest in one?
- ▶ Maybe, maybe not
 - ▶ It could be that the programs have a causal impact on student scores
 - ▶ But students that take test prep may be different from students that do not. They could vary in many dimensions: academic ability, college interest, as well as parental background
- ▶ With only **observational data** it is difficult to tell which story is true

Towards Causality

- ▶ In medicine, there is a long-established way to infer causality: the randomized control trial (RCT)

Towards Causality

- ▶ In medicine, there is a long-established way to infer causality: the randomized control trial (RCT)
- ▶ Imagine we are testing whether a given drug reduces blood pressure
- ▶ To study the effectiveness we recruit a group of participants

Towards Causality

- ▶ In medicine, there is a long-established way to infer causality: the randomized control trial (RCT)
- ▶ Imagine we are testing whether a given drug reduces blood pressure
- ▶ To study the effectiveness we recruit a group of participants
- ▶ Half of the participants are randomly placed into the **treatment** group and given the drug
- ▶ The other half are placed into the **control** or **placebo** group and given a placebo drug
- ▶ Now we can compare outcomes for the two groups over time to understand the impact of the drug

Randomization

- ▶ The key to a well-done experiment is **randomization**
- ▶ If who receives the treatment is completely random, then on average, the two groups will be very similar
- ▶ For example, in our test prep example, if we could randomly split students into two groups: test prep vs. no test prep, then we could evaluate the impacts of the program
- ▶ Since students are no longer self-selecting into the program, there should be little difference in the two groups in terms of academic ability, college interest, parental background

Experiments in Social Science

- ▶ Social scientists have borrowed from medicine to run experiments to answer causal questions
- ▶ Not all questions can be answered with an experiment, but if an experiment can be done, it is hard to find more convincing evidence
- ▶ Experiments have been particularly influential in development economics, with many projects aiming to develop policies to alleviate poverty in developing countries
 - ▶ In 2019, Abhijit Banerjee, Esther Duflo, and Michael Kremer won the Nobel Prize in Economics “for their experimental approach to alleviating global poverty.”
- ▶ We will be studying a number of social science experiments throughout this course

Our First Experiment

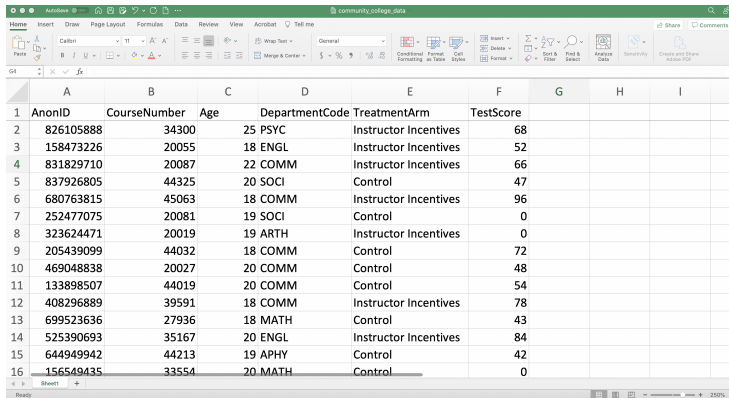
- ▶ Motivation: only 40 percent of community college students earn a college degree within 6 years (Shapiro et al. 2017)
- ▶ Question: Given poor outcomes, important to consider tools to improve performance. Is it possible financial incentives to instructors might improve teaching?

Our First Experiment

- ▶ Motivation: only 40 percent of community college students earn a college degree within 6 years (Shapiro et al. 2017)
- ▶ Question: Given poor outcomes, important to consider tools to improve performance. Is it possible financial incentives to instructors might improve teaching?
- ▶ In Brownback and Sadoff (2020) the authors run an experiment. At a community college in Indiana (Ivy Tech) some instructors (the treatment group) received \$50 for each student that passes the test (defined as above 70 percent), while the control group did not receive such an incentive
- ▶ Since on average the instructors will be similar in treatment and control, any difference in performance between students must be due to the treatment

Data Table = Observations*Variables

- ▶ Each row in a data table is an observation
- ▶ Each column is a variable



	A	B	C	D	E	F	G	H	I
1	AnonID	CourseNumber	Age	DepartmentCode	TreatmentArm	TestScore			
2	826105888	34300	25	PSYC	Instructor Incentives	68			
3	158473226	20055	18	ENGL	Instructor Incentives	52			
4	831829710	20087	22	COMM	Instructor Incentives	66			
5	837926805	44325	20	SOCI	Control	47			
6	680763815	45063	18	COMM	Instructor Incentives	96			
7	252477075	20081	19	SOCI	Control	0			
8	323624471	20019	19	ARTH	Instructor Incentives	0			
9	205439099	44032	18	COMM	Control	72			
10	469048838	20027	20	COMM	Control	48			
11	133898507	44019	20	COMM	Control	54			
12	408296889	39591	18	COMM	Instructor Incentives	78			
13	699523636	27936	18	MATH	Control	43			
14	525390693	35167	20	ENGL	Instructor Incentives	84			
15	644949942	44213	19	APHY	Control	42			
16	156549435	33554	20	MATH	Control	0			

The unit of observation

- ▶ Every time you open a new dataset, it is important to understand the unit of observation
- ▶ The unit of observation is the level at which the data is reported
- ▶ To discern it for yourself, think about what each row of the dataset represent
- ▶ Is each row an individual, a neighborhood, a country, a state?

What is the unit of observation?

E	F
Student ID	GPA
1	3.4
2	3.8

- ▶ Each row here is a student
- ▶ Therefore the observation level is a student

What is the unit of observation?

A	B	C
Student ID	Term	GPA
1	Fall	3
1	Winter	3.4
1	Spring	3.8
2	Fall	4
2	Winter	3.8
2	Spring	3.6

- ▶ Each row here is a student in a given term
- ▶ Therefore the observation level is a student-term

What is the unit of observation in our dataset

- ▶ **AnonID** is the identifier for a student
- ▶ **CourseNumber** is the identifier for a course
- ▶ Each row corresponds to an individual student within a given course
- ▶ Therefore the observation level is a student-course

Freezing the first row

- ▶ The first row contains the name of the variables
- ▶ As we scroll through or data, it is often helpful to **freeze** the first row so that it is always visible
 - ▶ This way you don't need to remember what each column corresponds to
- ▶ To freeze the first row go to the **View** tab and then click **Freeze Top Row**

Freezing the first row

The screenshot shows the Microsoft Excel interface with the 'View' tab selected. The 'Freeze Top Row' button is highlighted with a red box. The spreadsheet data is as follows:

	A	B	C	D	E	F	G	H	I
1	AnonID	CourseNumber	Age	DepartmentCode	TreatmentArm	TestScore			
2	826105888	34300	25	PSYC	Instructor Incentives	68			
3	158473226	20055	18	ENGL	Instructor Incentives	52			
4	831829710	20087	22	COMM	Instructor Incentives	66			
5	837926805	44325	20	SOCI	Control	47			
6	680763815	45063	18	COMM	Instructor Incentives	96			
7	252477075	20081	19	SOCI	Control	0			
8	323624471	20019	19	ARTH	Instructor Incentives	0			
9	205439099	44032	18	COMM	Control	72			
10	469048838	20027	20	COMM	Control	48			
11	133898507	44019	20	COMM	Control	54			
12	408296889	39591	18	COMM	Instructor Incentives	78			
13	699523636	27936	18	MATH	Control	43			
14	525390693	35167	20	ENGL	Instructor Incentives	84			
15	644949942	44213	19	APHY	Control	42			
16	156549435	33554	20	MATH	Control	0			

Summary of Data

- ▶ This is the actual data that comes the experiment run by Brownback and Sadoff (2020)
- ▶ Our goal is to determine if test scores depend on the Treatment Arm of the student
 - ▶ In other words, based on whether the instructor has financial incentives to improve test scores or not
- ▶ In order to learn whether the treatment is effective, we need to learn how to use **functions** in Excel